

PRS821S PROGRAMMING FOR SECURITY PERSONNEL, SEM 1, 2026 (WEEK 12)

THE WHITE-BOX, BLACK-BOX AND ZOO ATTACKS

When we talk about adversarial attacks on machine learning models, **white-box attacks** are often discussed first. In a white-box scenario, the attacker has full access to the internal structure of the target model, including its weights and architecture. This allows for highly effective attacks but is less realistic in real-world scenarios where such information is unavailable.

White box attacks involve **full knowledge** of a system's internal structure, code, or model parameters to craft precise, highly effective threats. Black box attacks assume **no prior knowledge**, relying on input-output probing to uncover vulnerabilities, simulating realistic, external adversarial actions. White box is precise but resource-heavy; black box is realistic but often less efficient.

On the other hand, **black-box attacks** simulate more practical threats. In these attacks, only the input and output of the model are accessible, but the inner workings remain hidden. This is where the **ZOO (Zeroth Order Optimization)** attack method steps in. What makes ZOO unique is that it allows attackers to generate adversarial examples without relying on a substitute model. With Zoo attacks, attackers don't have access to a model's internal architecture.

Zeroth Order Optimization (ZOO)

Zeroth Order Optimization (ZOO) is a method that allows for gradient estimation even when the model's internal details aren't available (as in black-box attacks). Instead of relying on backpropagation, which requires knowledge of the model, ZOO directly estimates gradients from the input-output relationship of the model. This makes it incredibly powerful for attacking models in black-box settings.

Zeroth Order Optimisation (ZOO) based attacks directly estimate the gradients of the targeted models for generating adversarial examples.

White Box Attacks (Full Knowledge)

- **Definition:** The attacker has complete access to the system, including source code, architecture, and network diagrams.
- **Methodology:** Uses deep analysis of code and configuration to find vulnerabilities, such as SQL injection, buffer overflows, or weak encryption keys.
- **Machine Learning (ML) Context:** Attacks involve accessing model gradients, architecture, and training data to craft optimized adversarial examples.
- **Advantages:** Provides comprehensive security coverage and identifies weaknesses in internal logic.
- **Disadvantages:** Time-consuming and requires significant resources and expertise.

Black Box Attacks (No Knowledge)

- **Definition:** The attacker (or tester) has zero knowledge of the internal workings, similar to an external, unprivileged threat actor.
- **Methodology:** Operates purely through input manipulation and output observation, focusing on functional testing.
- **Machine Learning (ML) Context:** Uses querying techniques (e.g., SpecterOps) to observe labels or scores, often using surrogate models to create transferable adversarial examples.
- **Advantages:** Simulates real-world, unexpected attacks and highlights vulnerabilities visible to outsiders.
- **Disadvantages:** May miss deep structural flaws and can be inefficient.

Comparison Table

Feature	White Box Attack	Black Box Attack
Knowledge	Full (Source code, design)	None (External entity)
Effort	High	Moderate/High
Efficiency	High (Targeted)	Low (Guesswork)
Scope	Comprehensive	Functional
Used For	Insider threats, audit	External breaches